API Documentation and Postman Testing Report

Overview

The purpose of this task was to verify the functionality of backend APIs, perform API testing using Postman, validate request and response handling, and prepare API documentation for future development and integration.

The APIs tested in this phase include:

- User Profile APIs
- Product APIs
- User Measurements APIs
- Virtual Try-On APIs

The testing process ensured that all endpoints function correctly, return expected responses, handle invalid requests appropriately, and maintain proper data flow between the frontend, backend, and database.

API Verification Process

The following verification steps were performed for each API:

1. Endpoint Verification

- Verified that API routes are correctly configured.
- Checked route accessibility through Postman.
- Confirmed proper HTTP methods are used.

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2. Request Validation

- Tested APIs using valid request data.
- Tested APIs using missing or invalid fields.
- Verified validation and error handling.

3. Database Verification

- Confirmed data insertion into MongoDB collections.
- Verified update and retrieval operations.
- Checked consistency between request data and stored records.

4. Response Verification

- Confirmed successful response status codes.
- Verified response body structure.
- Ensured meaningful error messages are returned.

5. Authentication Verification

- Tested protected routes using JWT tokens.
- Verified unauthorized access restrictions.
- Confirmed token validation functionality.

Postman Testing Procedure

The following steps were used while testing APIs in Postman:

Step 1: Configure Request

- Select HTTP Method (GET, POST, PUT, DELETE)
- Enter API Endpoint URL
- Add Authorization Token (if required)

Step 2: Add Request Data

For POST and PUT requests:

- Open Body tab
- Select Raw
- Choose JSON format
- Enter request payload

Step 3: Send Request

- Click Send
- Observe API response

Step 4: Validate Results

Verify:

- Status code
- Response body
- Database changes
- Error handling

Required Postman Screenshots

The following screenshots were captured for documentation purposes:

Successful Requests

- GET request success
- POST request success
- PUT request success
- DELETE request success

Error Handling

- Missing required fields
- Invalid input data
- Unauthorized access
- Resource not found

Authentication

- Valid JWT token access
- Invalid token access
- No token access

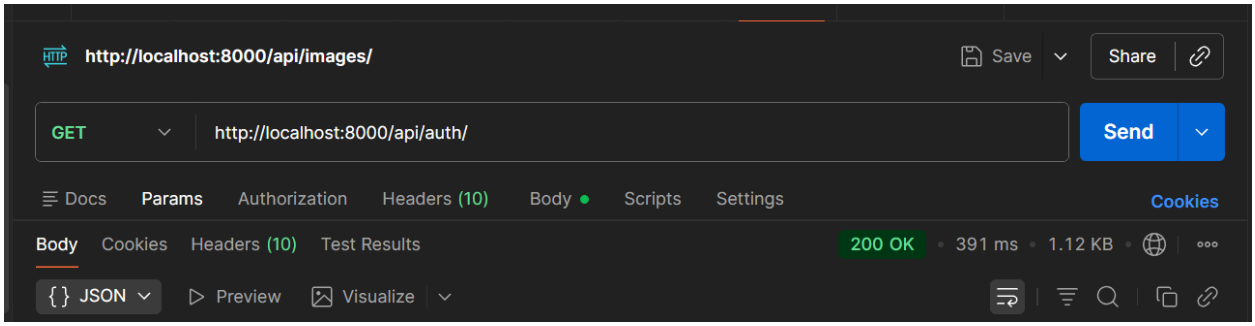
User Profile API Documentation

Purpose

The User Profile APIs manage user profile information, preferences, avatar data, and profile updates.

Get User Profile

Endpoint



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Authentication

Not Required

Success Response


```
        {
            "profileImage": "",
            "avatarImage": "",
            "bodyScanImage": "",
            "_id": "6a0ebaecf4dba41b08178721",
            "name": "John Doe",
            "email": "john@example.com",
            "password": "$2b$10$c2EBLSBv6bG83Cr.nAdKTOGbT9j4h6GCTJ/63EPwOHMLvsdisCnBC",
            "__v": 0
        },
        {
            "_id": "6a154c4b57b9020f4efcf0b6",
            "name": "testuser",
            "email": "test@example.com",
            "password": "$2b$10$THcex/aDvj.L.CzfcBBIX./wmGcBYMf0y5HCS1sgcCu6v1t3AzVBC",
            "profileImage": "",
            "avatarImage": "",
            "bodyScanImage": "",
            "createdAt": "2026-05-26T07:31:23.931Z",
            "updatedAt": "2026-05-26T07:31:23.931Z",
            "__v": 0
```

```
            "bodyScanImage": "",
            "createdAt": "2026-05-26T07:31:23.931Z",
            "updatedAt": "2026-05-26T07:31:23.931Z",
            "__v": 0
        },
        {
            "_id": "6a15527a803283b5231f8d30",
            "name": "jone",
            "email": "jone@gmail.com",
            "password": "$2b$10$K.ZFUWnpTYuCwaRnfBBWh.kCHHoCTr8nX9O7.lo0xoskoQjnS006O",
            "avatarImage": "https://res.cloudinary.com/dpf9q6s8b/image/upload/v1779798111/raritone/
                avatar/bqgujosul5vdpkdjm7oq.png",
            "bodyScanImage": "",
            "createdAt": "2026-05-26T07:57:46.621Z",
            "updatedAt": "2026-05-26T12:21:52.236Z",
            "__v": 0,
            "profileImage": "https://res.cloudinary.com/dpf9q6s8b/image/upload/v1779798084/raritone/
                profile/xljtupnlhhh1fiwtcv2x.png",
            "profileImagePublicId": "raritone/profile/vqvhqjedpbojvyq7k9nj"
        }
    ]
}
```


Expected Result

Returns complete user profile information.

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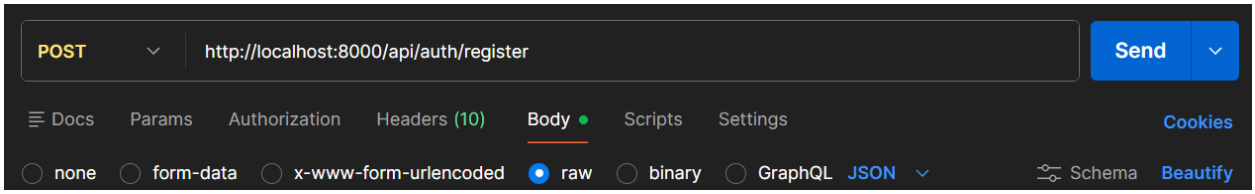
Post User Profile

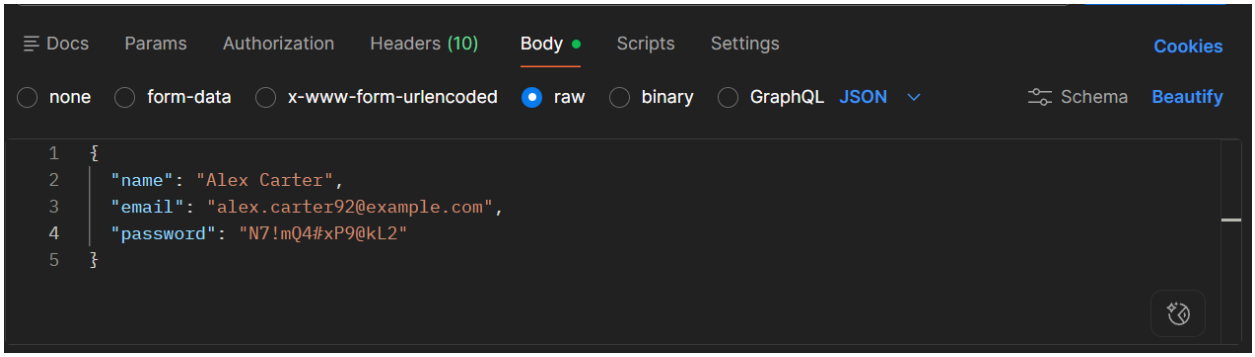
Endpoint



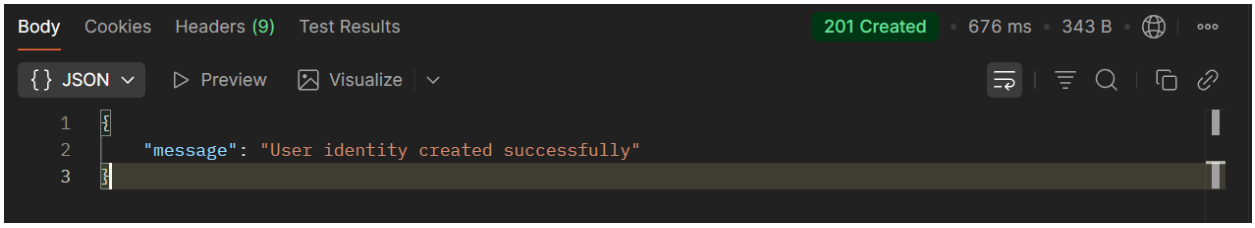
Authentication

Required

Request Body



Success Response



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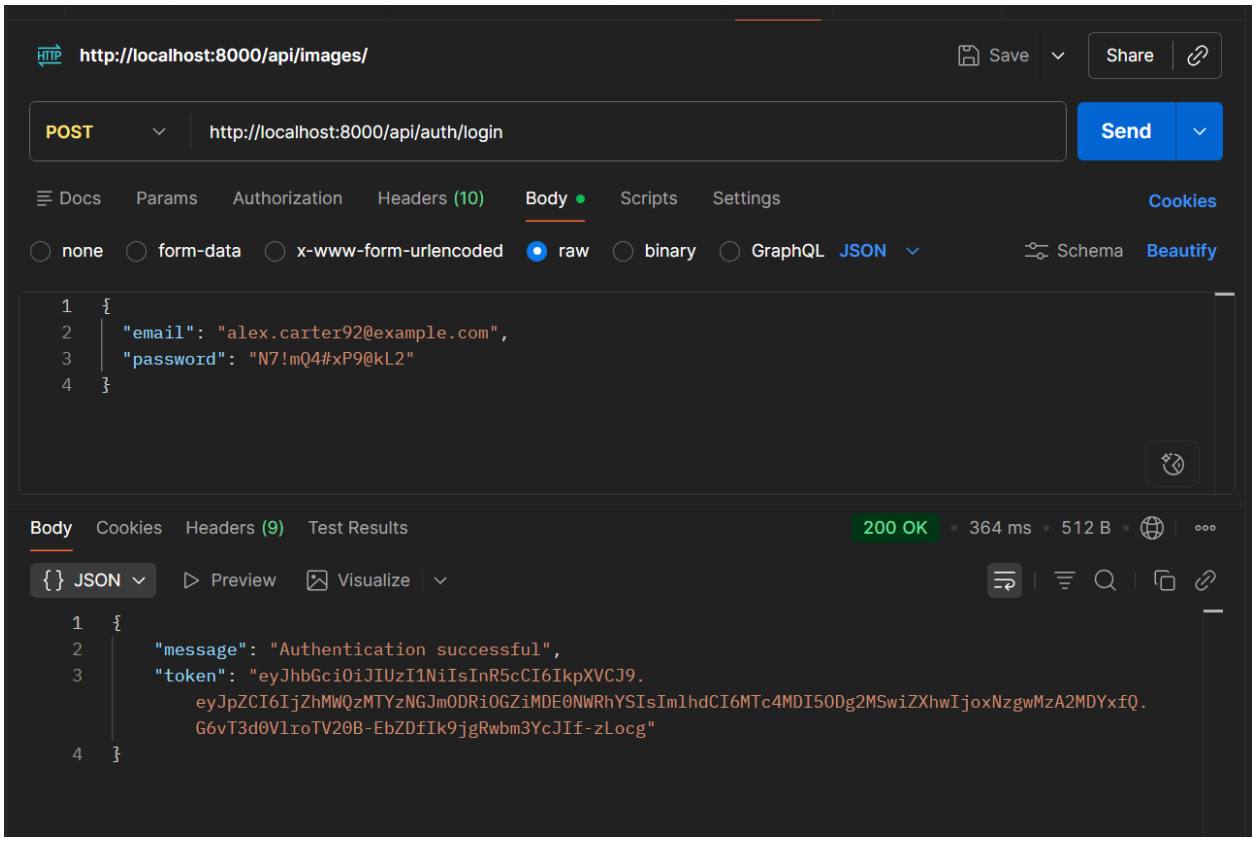
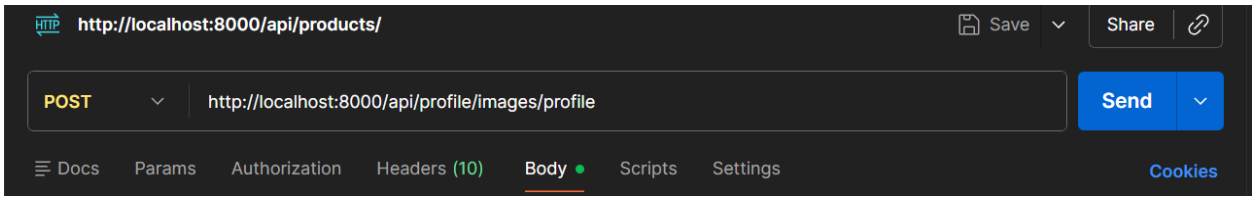
Post User Profile



Post Profile image

Endpoint

Post / api/profile/images/profile



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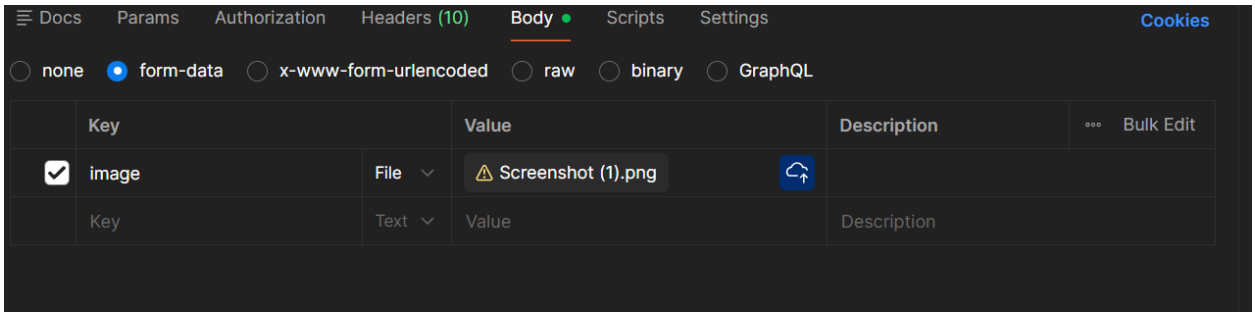
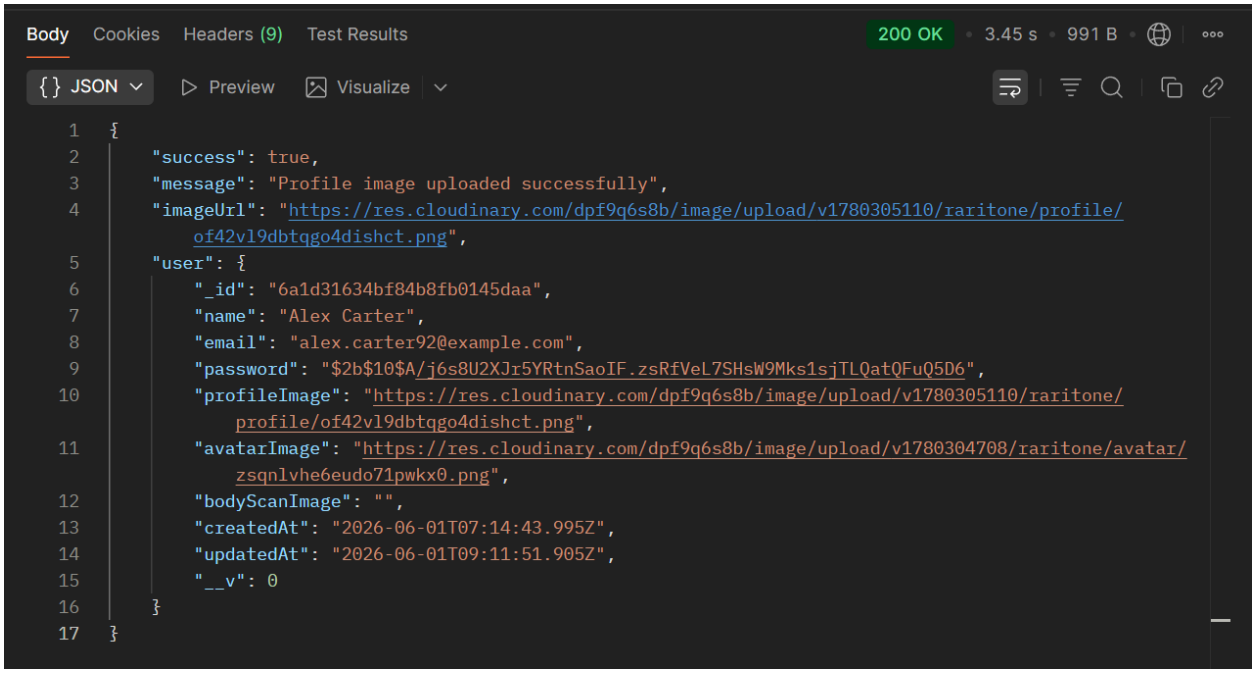
Authentication

Required

Request Body



Success Response

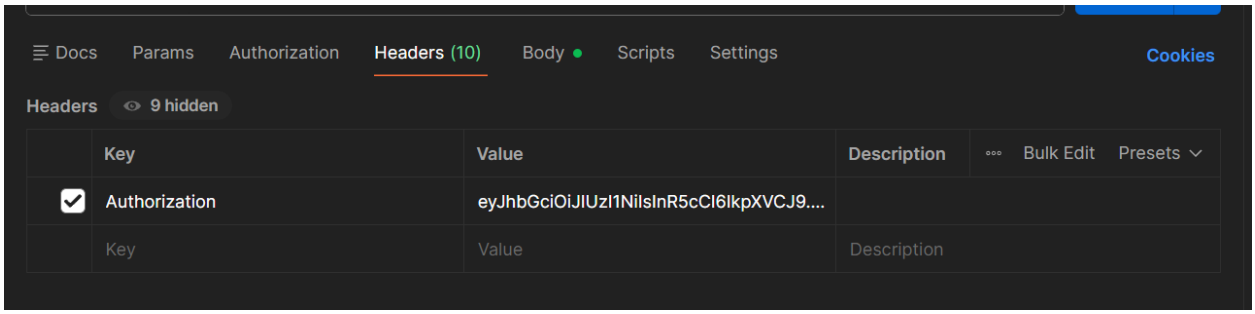


Post Profile avatar

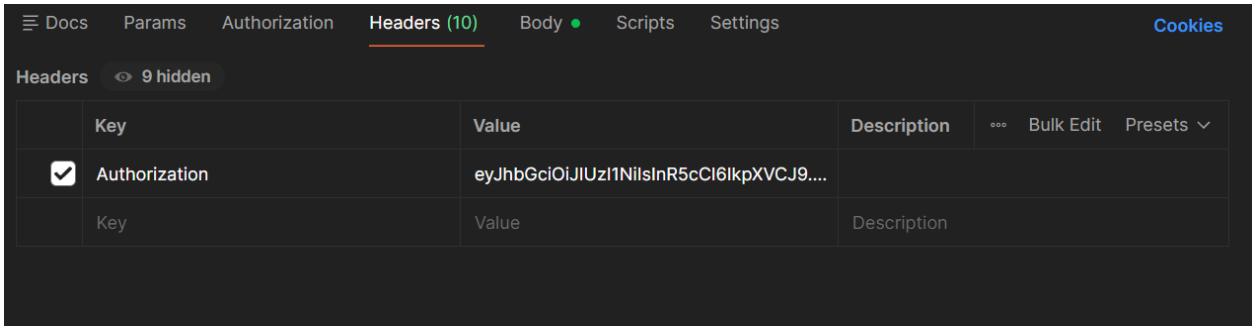
Endpoint

Post /api/profile/images/avatar

Authentication



Request Body



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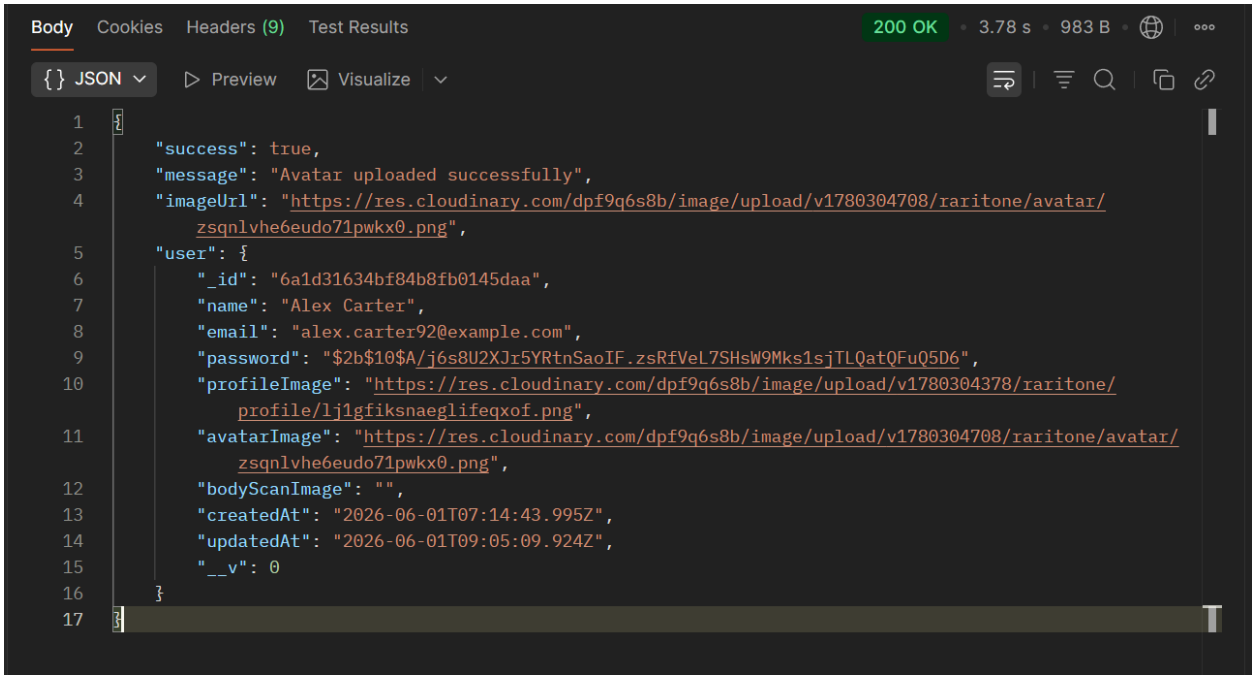
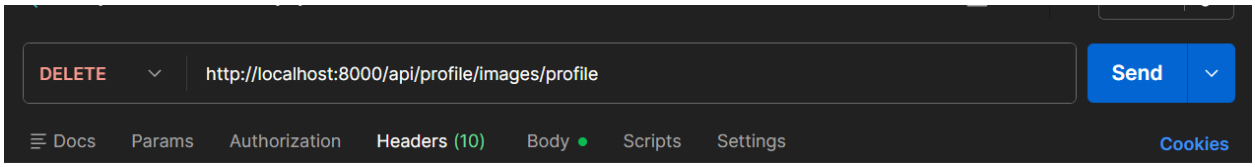
Success Response



Delete profile

Endpoint

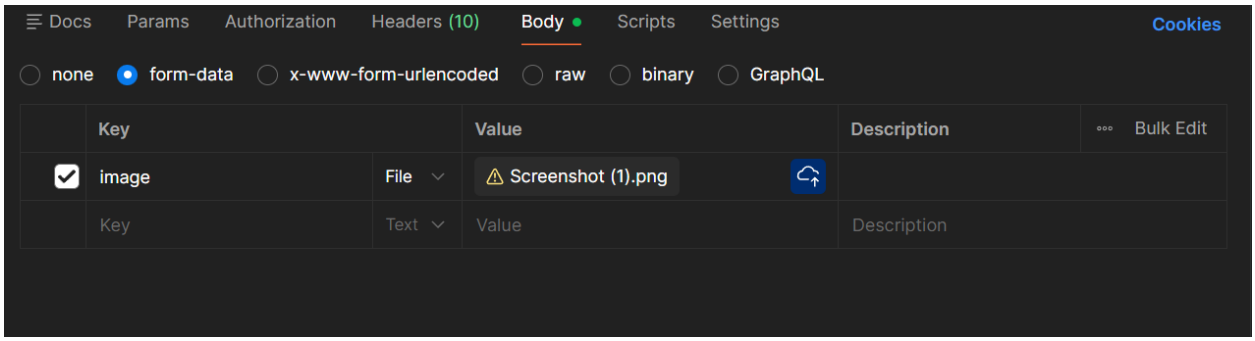


Authentication

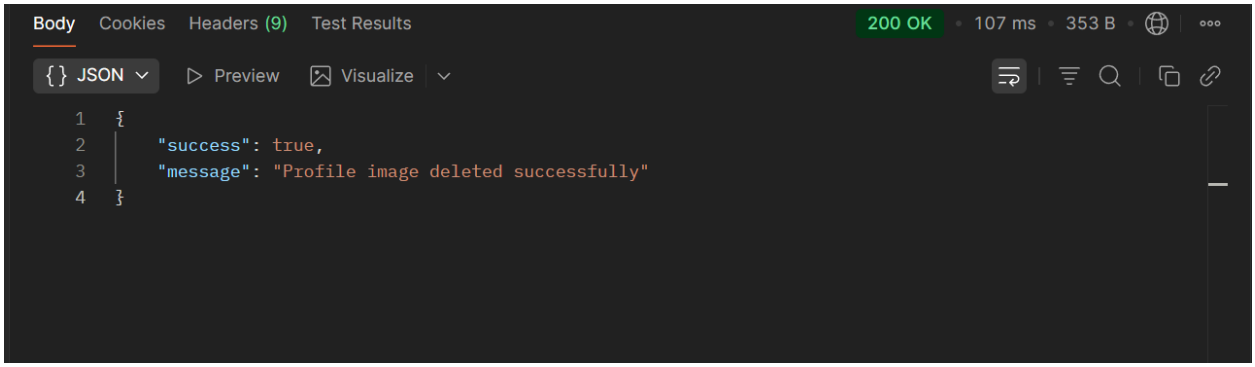
Required

Request Body

Required



Success Response



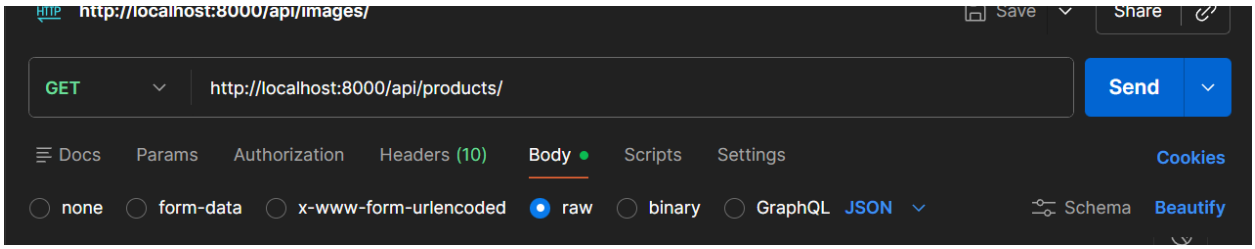
Product API Documentation

Purpose

Product APIs manage product listing, retrieval, creation, updating, and deletion

Get All Products

Endpoint



Authentication

Not Required

Request Body

Not Required

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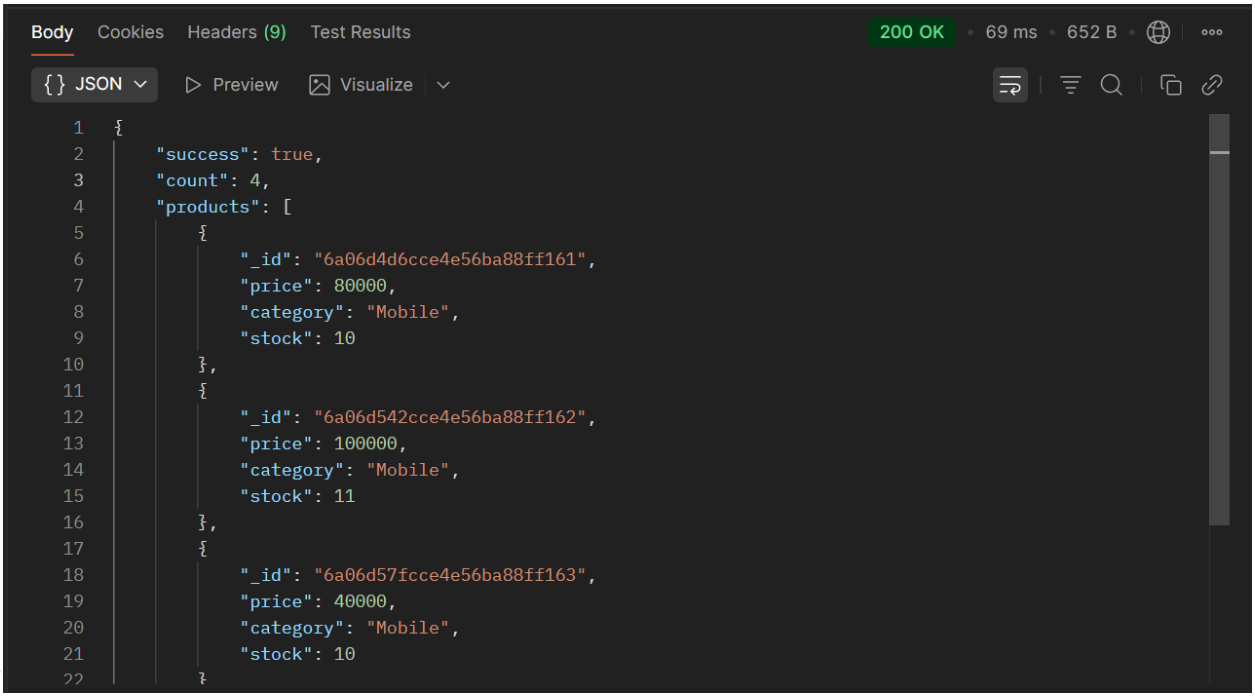
Success Response

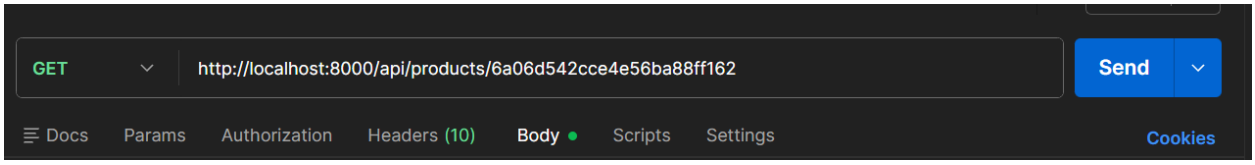


{

Get Single Product

Endpoint

GET /api/products/:id



Authentication and Request body

Not Required

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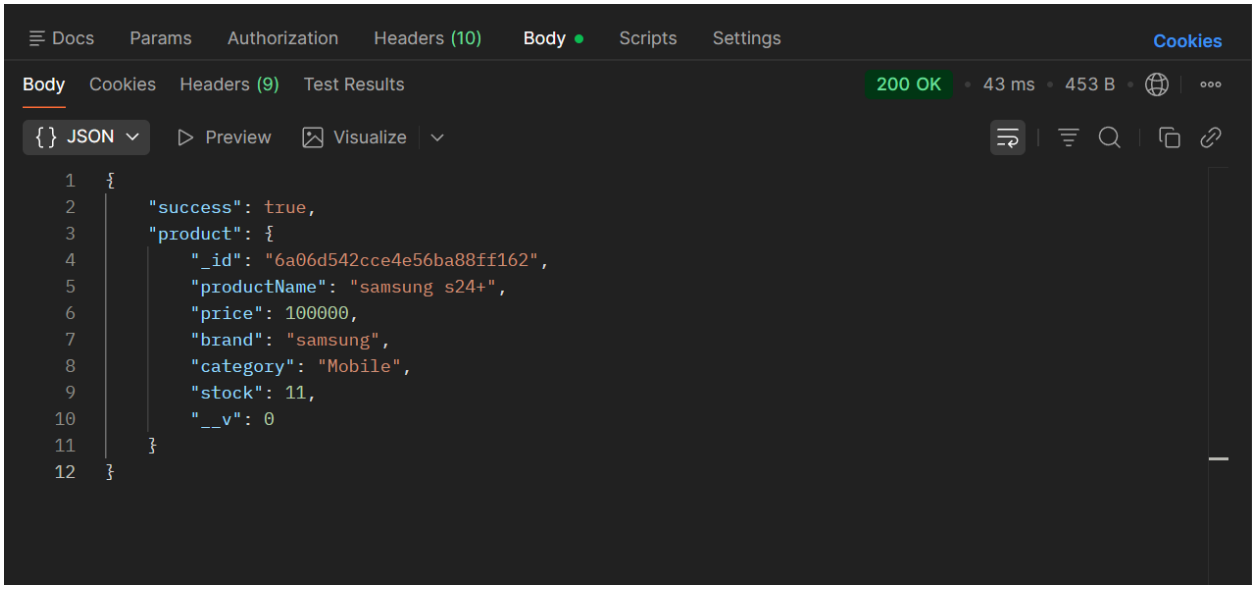
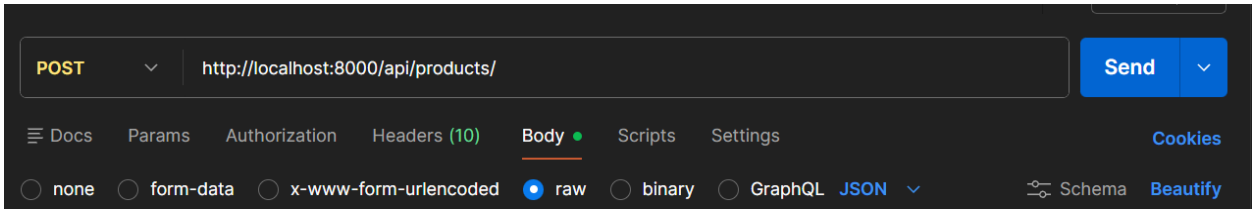
Success Response



Add Product

Endpoint

POST /api/products/



Authentication

Required

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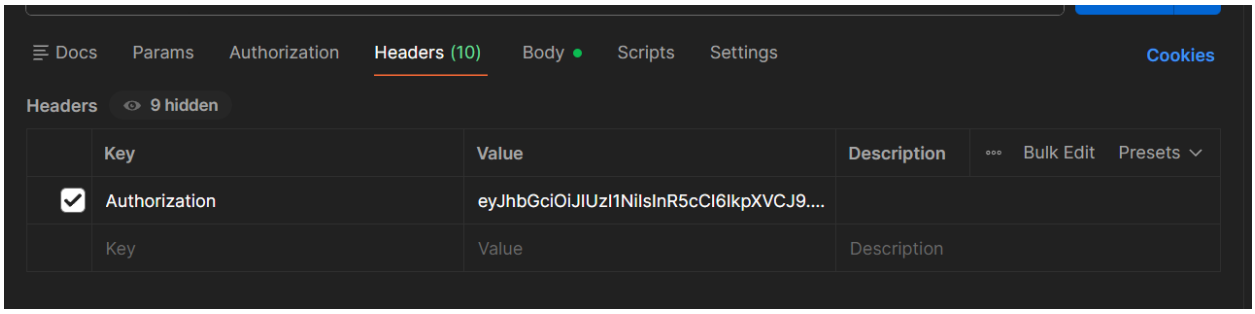
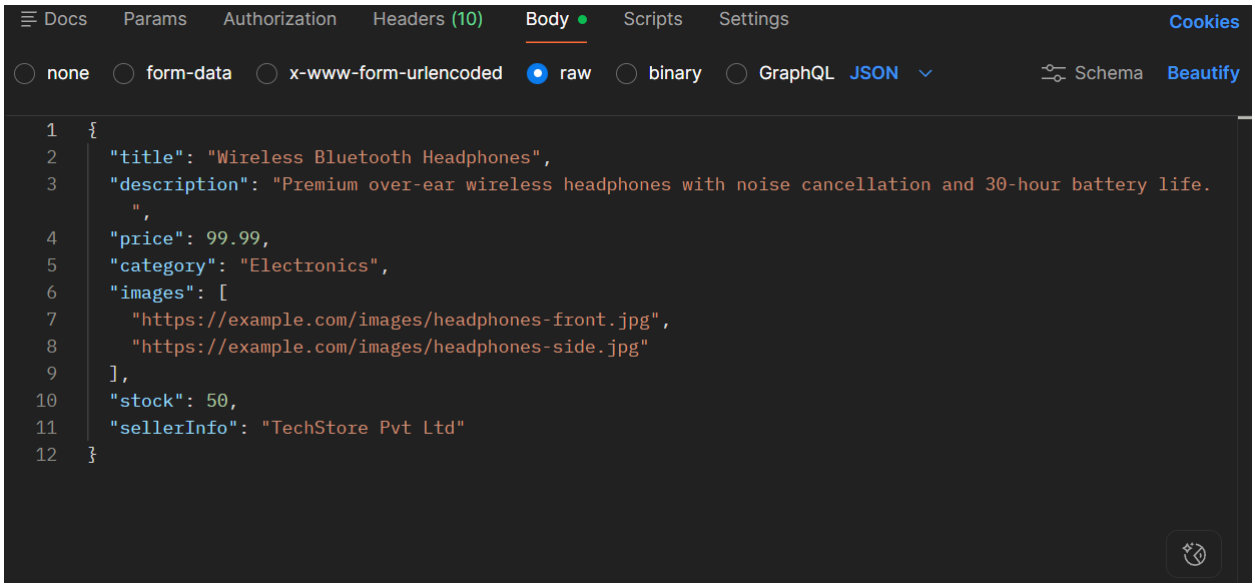
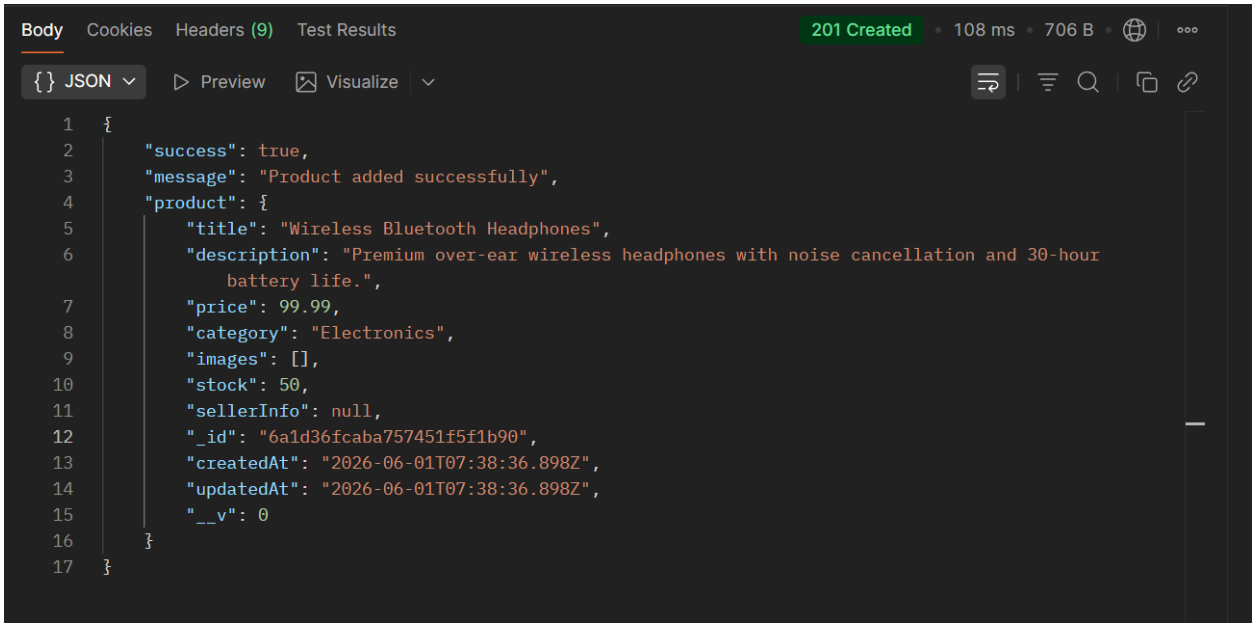


Request Body



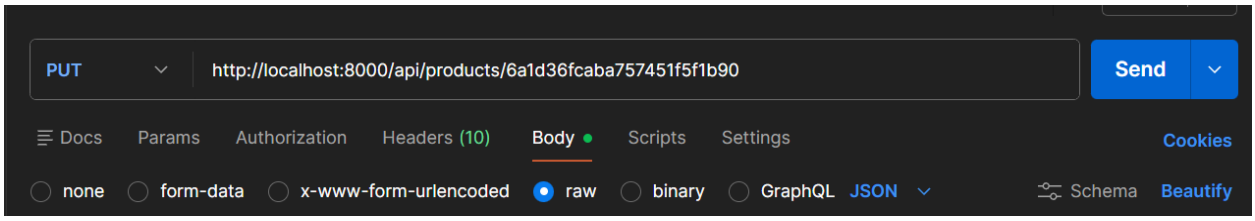
Success Response



Update Product

Endpoint

Put /api/products/:id

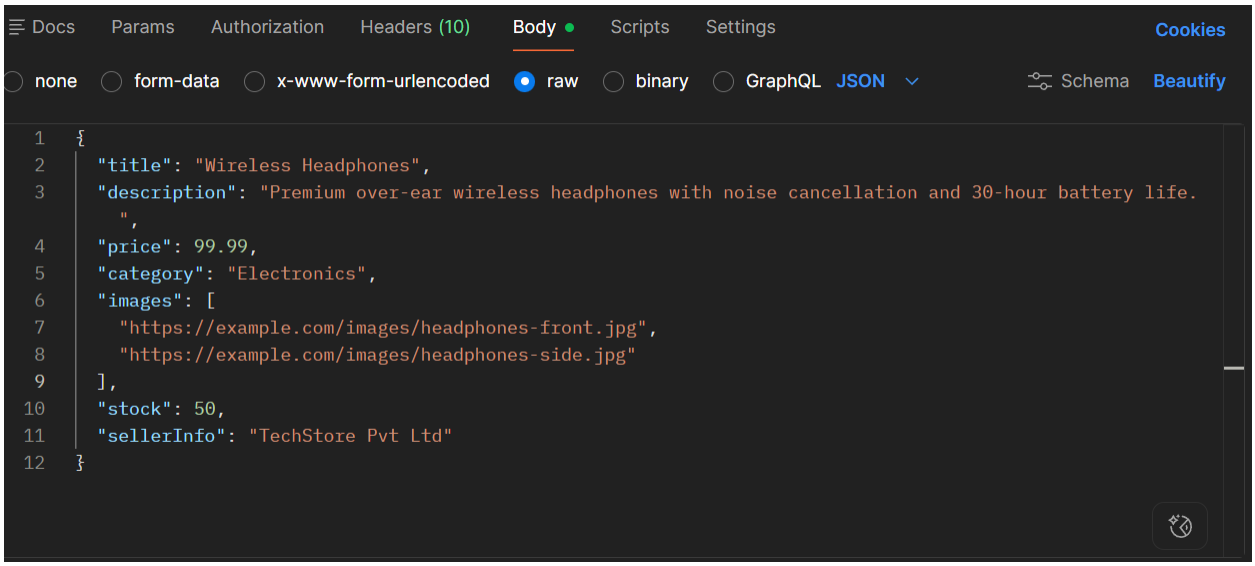


Authentication

Required

Request Body

Required



Success Response

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User Measurements API Documentation

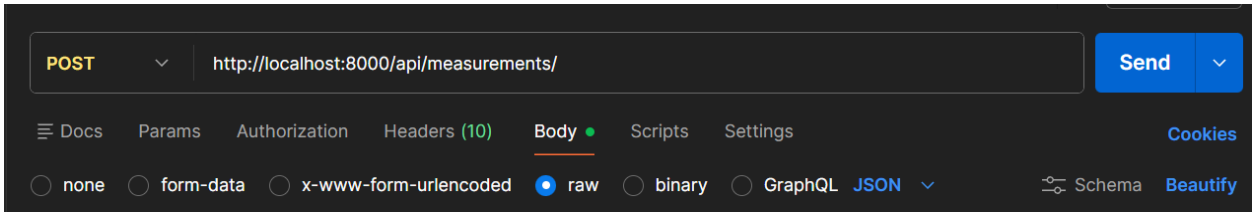
Purpose

Stores user body measurements to improve AI-based virtual try-on accuracy and size recommendations.

Add Measurements

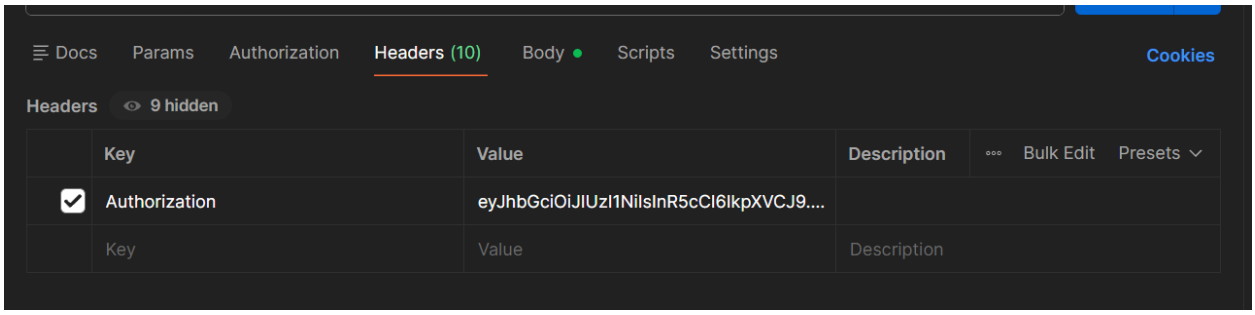
Endpoint

POST /api/measurements/



Authentication

Required



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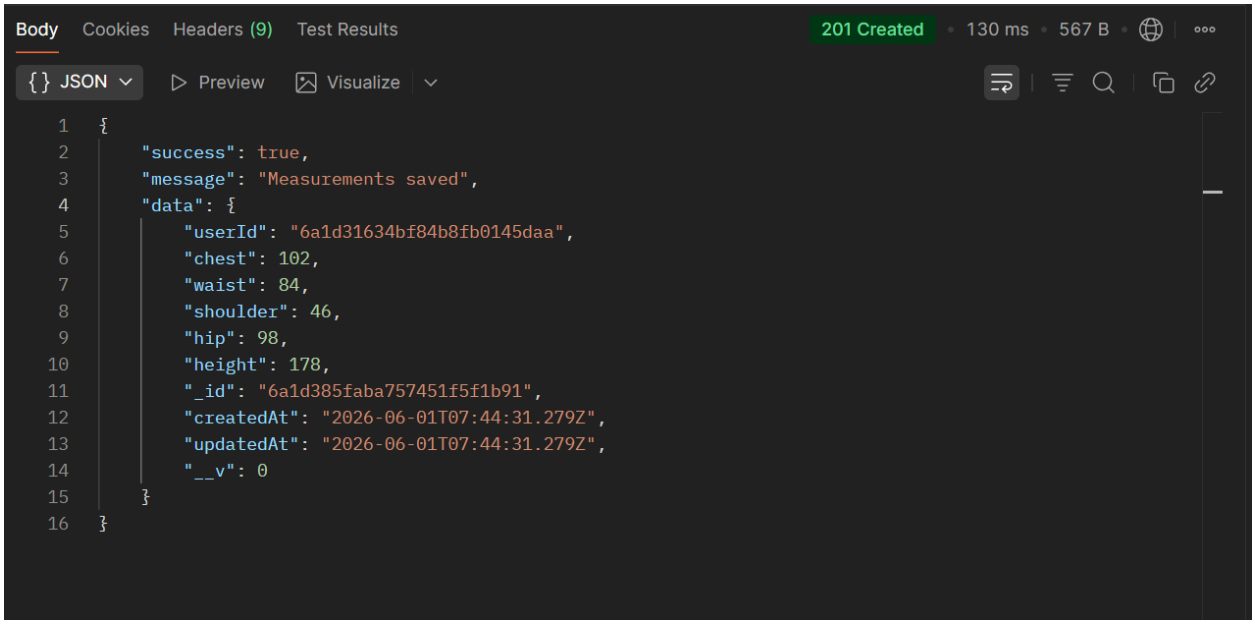
Request Body

```
{
  "userId": "665f1a2b3c4d5e6f7890abcd",
  "chest": 102,
  "waist": 84,
  "shoulder": 46,
  "hip": 98,
  "height": 178
}
```

Success Response



Get Measurements

Endpoint

GET /api/measurements/

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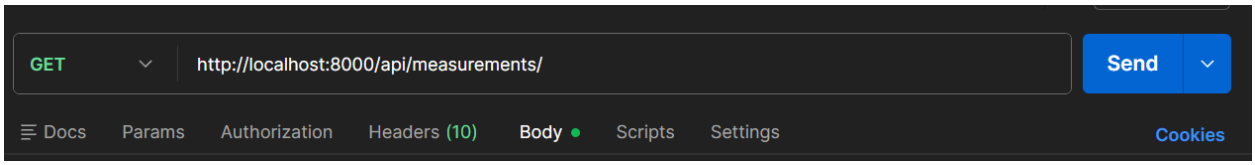
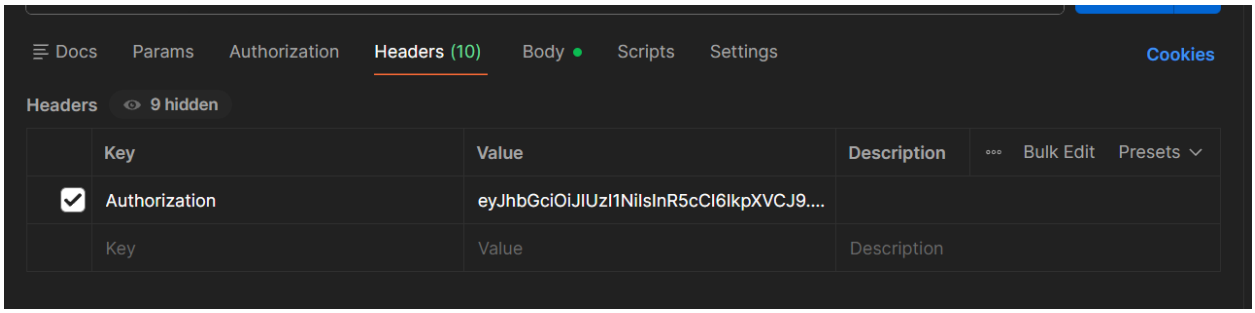


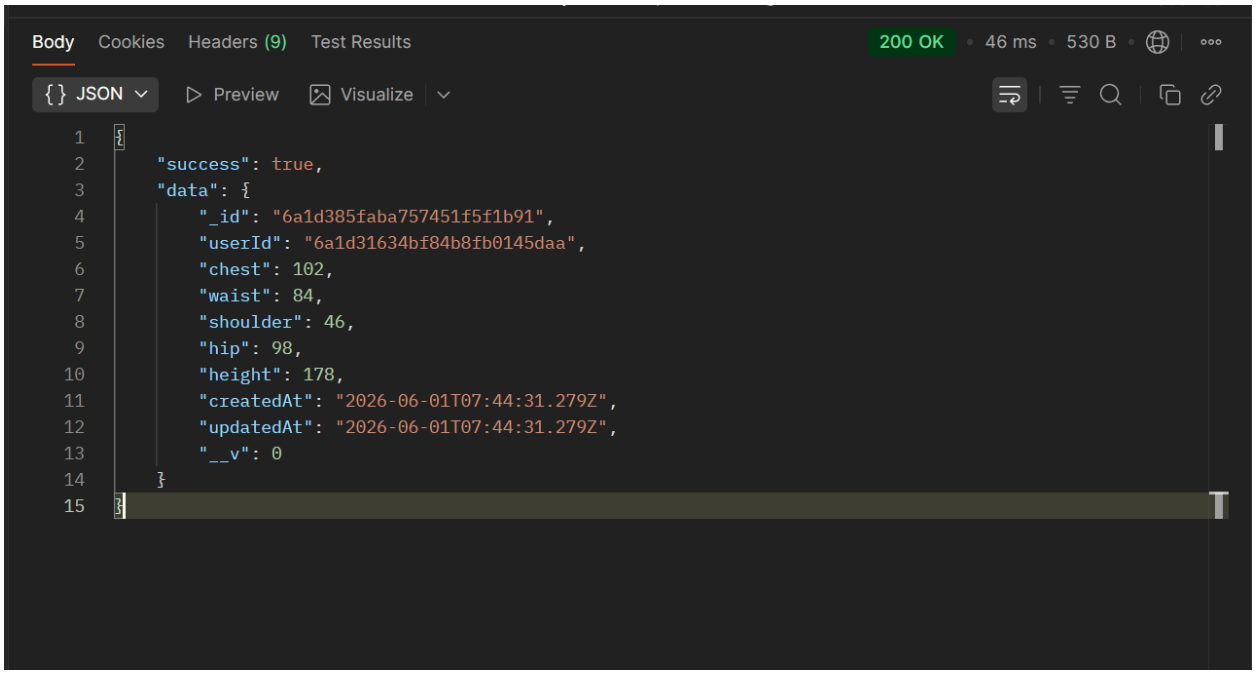
Authentication

Required



Request Body : Not Required

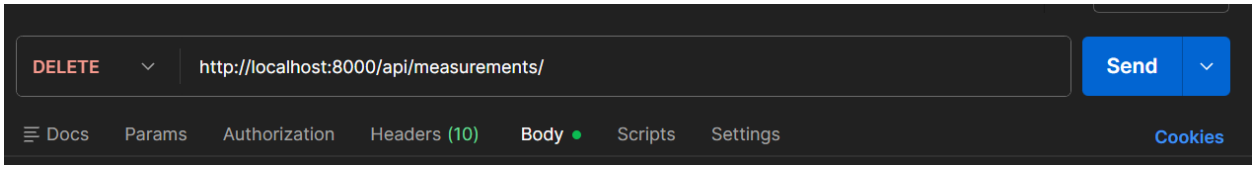
Success Response



Delete Measurements

Endpoint

delete/api/measurements/



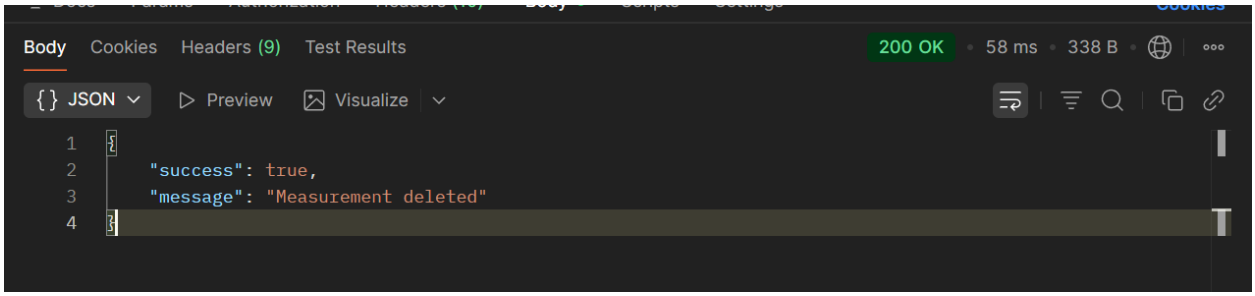
Authentication

Not Required

Request Body

Not Required

Success Response



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Virtual Try-On API Documentation

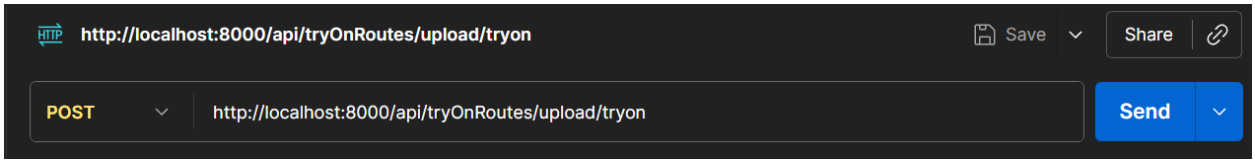
Purpose

Virtual Try-On APIs allow users to upload images and generate AI-powered try-on previews.

Upload User Image

Endpoint

POST /api/tryon/upload/tryon



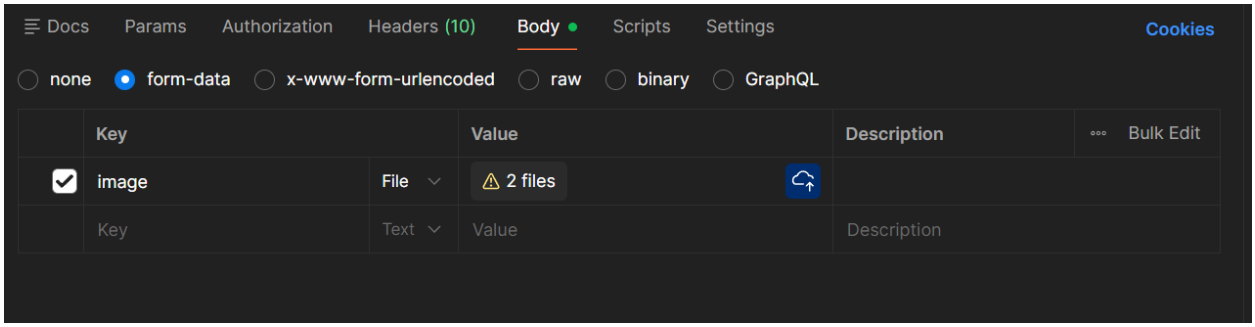
Authentication

Required

Request Body

Form Data

`image: file`



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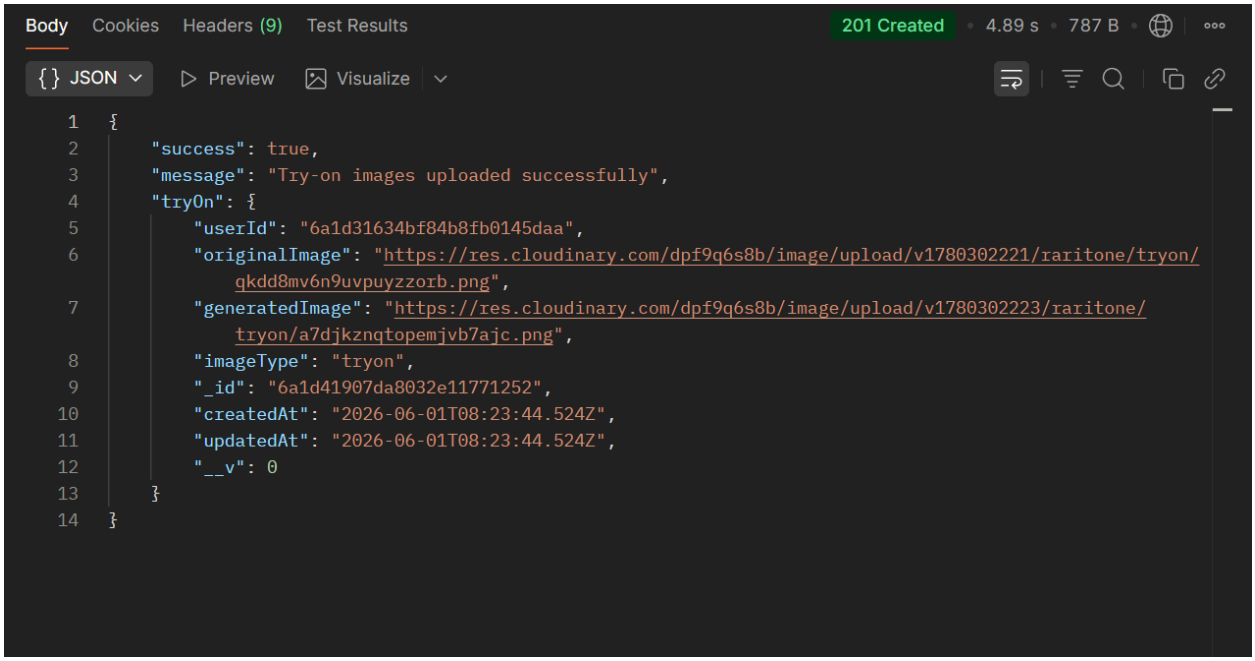
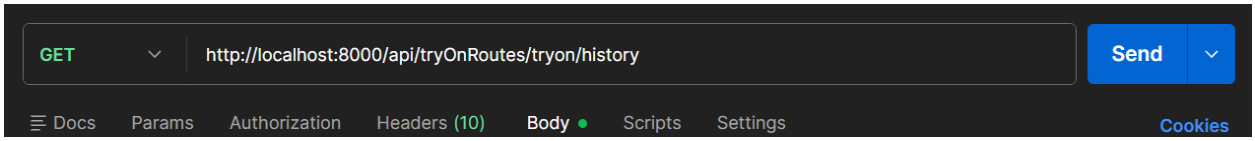
Success Response



Get Virtual Try-On

Endpoint

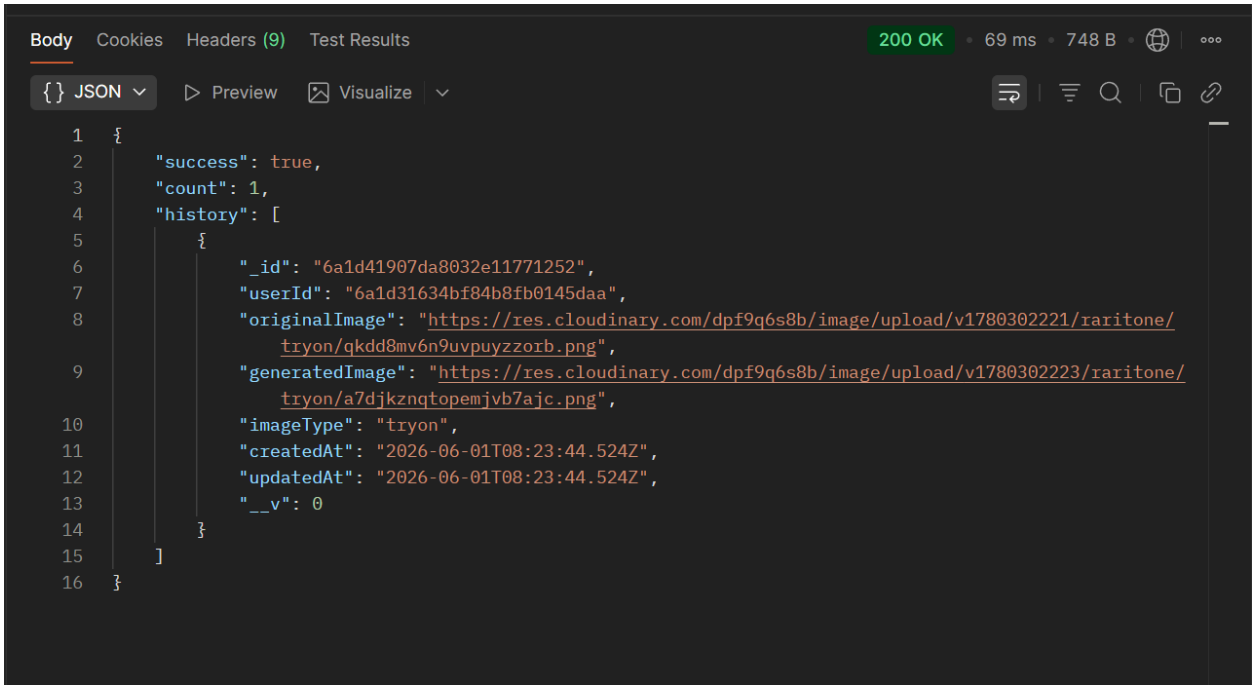
POST / api/tryOnRoutes/tryon/history



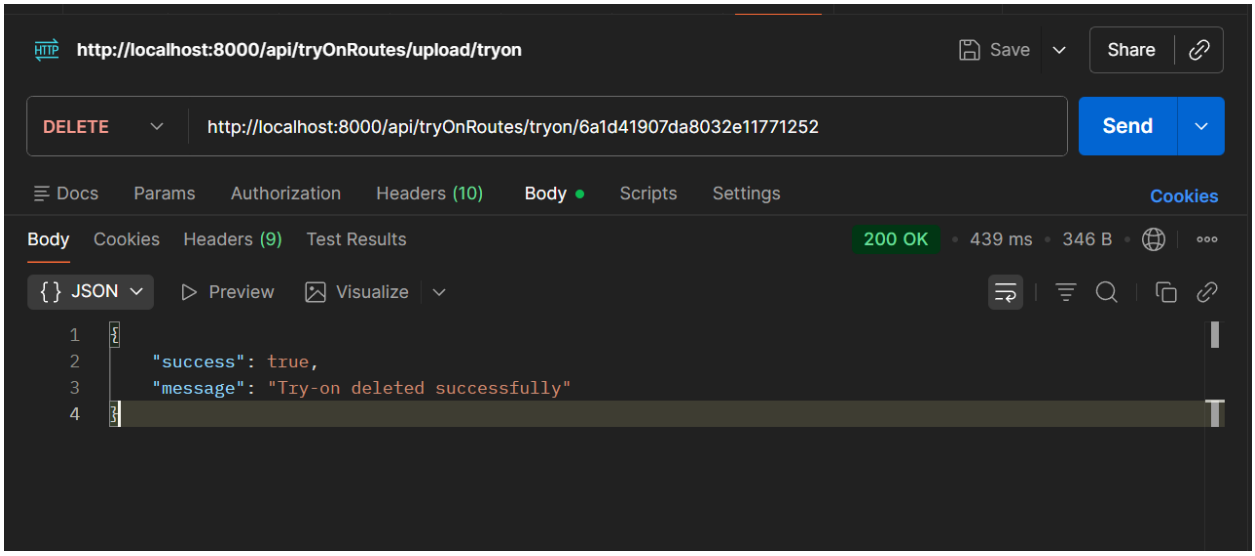
Authentication

Required

Success Response



Delete request Virtual Try-On:



Validation Error Testing

The following validation scenarios were tested:

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Missing Required Fields

Example:

```
{
  "height": 175
}
```

Expected Response:

```
{
  "success": false,
  "message": "Required fields missing"
}
```

Invalid Data Type

Example:

```
{
  "height": "abc"
}
```

Expected Response:

```
{
  "success": false,
  "message": "Invalid data type"
}
```

Unauthorized Access

Expected Response:

```
{
  "success": false,
  "message": "Unauthorized access"
}
```

API Verification Summary

The APIs were verified for:

- Route functionality
- Request validation
- Response handling
- Database integration
- Authentication checks
- Error handling

Testing confirmed that the APIs are functioning correctly and are ready for frontend integration and future deployment.

Deliverables

Completed Deliverables:

✅ API Documentation

✅ API Verification

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- ✅ Postman Testing
- ✅ Success Response Testing
- ✅ Validation Error Testing
- ✅ Authentication Testing
- ✅ Postman Screenshots Collection

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